

Aoao (Olivia) Xu

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Research Profile

Finance graduate with research interests in empirical asset pricing, sustainable finance, financial machine learning, and computational analysis of corporate disclosures. My academic training combines finance, econometrics, portfolio analysis, machine learning, and natural language processing. I have published two peer-reviewed articles and completed research-oriented projects involving cross-sectional return prediction, portfolio construction, panel-data modelling, sentiment analysis, topic modelling, and large-language-model-based extraction from SEC filings. My proposed doctoral research examines whether AI-enabled measurement of ESG disclosure credibility can provide incremental information for asset pricing and firm risk.

Research Interests

Empirical asset pricing; sustainable finance; ESG disclosure credibility and greenwashing; financial machine learning; financial textual analysis; natural language processing and large language models; portfolio construction; information asymmetry; and financial econometrics.

Education

Master of Finance

Jul 2024 – Dec 2025

The University of Melbourne, Melbourne, Australia

- GPA: 3.63/4.0 | WAM: 76.50%.
- Completed the University's selective, small-cohort Master of Finance.
- Relevant coursework: Applied Machine Learning in Finance — 82 (H1); Quantitative Analysis of Finance I — 80 (H1).

Bachelor of Applied Finance (First Class)

Jul 2022 – Jun 2024

Macquarie University, Sydney, Australia

- GPA: 4.0/4.0 | WAM: 83.81%.
- Relevant coursework: Stochastic Methods in Applied Finance — 96 (HD); Financial Modelling — 93 (HD); Quantitative Methods in Economics, Business and Finance — 86 (HD).

Bachelor of Finance

Sep 2020 – Jun 2022

Jiangsu University, Jiangsu, China

- GPA: 3.95/5.0 | Overall grade: 89.50%.
- Ranked first in the Finance major for four consecutive semesters during 2020–2022.

Publications

Xu, A. (Olivia). "Research on the Application of Digital RMB to Address the Monopoly and Privacy Issues of Third-Party Payments." *International Journal of Management Sciences and Business Research*, Vol. 12, No. 5, 2023. DOI: [10.5281/zenodo.8013569](https://doi.org/10.5281/zenodo.8013569). **First author**.

Xu, A. et al.. "Research on Household Emergency Supplies Storage from the Theory of Planned Behavior and Intention–Behavior Gap in the Context of COVID-19." *Frontiers in Psychology*, 2023. DOI: [10.3389/fpsyg.2022.1069843](https://doi.org/10.3389/fpsyg.2022.1069843). **Co-author**.

Research Experience

Machine Learning-Based Return Prediction and Portfolio Construction

Oct 2025

The University of Melbourne — Applied Machine Learning in Finance

- Developed a machine-learning framework for cross-sectional return prediction using financial factor data and ridge regression to estimate future stock returns.
- Implemented rolling-window model estimation and compared predictive performance across multiple regularisation parameters.
- Converted predicted returns into portfolio weights and evaluated out-of-sample portfolio performance using realised returns and annualised Sharpe ratios.
- Identified the regularisation specification associated with the strongest out-of-sample risk-adjusted performance.

Financial NLP and Large-Language-Model Analysis of Corporate Disclosures

Oct 2025

The University of Melbourne — Applied Machine Learning in Finance

- Developed an end-to-end financial NLP pipeline using Google Gemini to analyse unstructured corporate disclosures from SEC Form 10-K filings.
- Applied prompt engineering and LLM-based information extraction to identify firms' AI-related disclosures and convert

qualitative text into structured variables for quantitative analysis.

- Designed a sentiment-classification framework categorising firms' AI-related outlook as Bullish, Neutral, or Bearish, accompanied by model-generated rationales supporting each classification.
- Transformed extracted textual information into investment indicators for analysing corporate AI exposure and its potential financial relevance.

Financial Data Analysis and Textual Analytics of Corporate Annual Reports

May 2025

The University of Melbourne — Data Analysis for Finance

- Conducted an empirical firm-level analysis combining stock performance, rolling CAPM beta estimation, firm characteristics, and panel-regression modelling.
- Applied financial textual-analysis methods to corporate annual reports, including sentiment analysis and Latent Dirichlet Allocation topic modelling, to examine disclosure tone and thematic content.
- Performed data cleaning, variable construction, and winsorisation, and compared alternative fixed-effects specifications.
- Used firm-level clustered standard errors to improve the reliability of panel-data statistical inference.

Factor Investing and Portfolio Analysis

May 2025

The University of Melbourne — Applied Investment Management

- Constructed characteristic-sorted long-short factor portfolios using Python and Australian equity data.
- Computed factor returns and risk-adjusted performance and benchmarked results against established academic factors, including the Jensen, Kelly, and Pedersen factor library.
- Interpreted differences between constructed and benchmark factors in relation to portfolio design, data coverage, and economic mechanisms.

Professional Experience

Financial Data Operations Analyst

May 2026 – Present

State Street, Hangzhou, China

- Reconcile transaction, position, and portfolio data for institutional investment clients across internal and external systems.
- Investigate reconciliation breaks and data discrepancies, identify root causes, and support timely resolution of post-trade issues.
- Perform data validation and exception monitoring to improve financial data accuracy and reduce operational risk.

Financial Product Sales Intern

Feb – Apr 2025

TMGM Pty Ltd, Melbourne, Australia

- Conducted market research and prospect identification for financial products and maintained structured client and performance records.
- Exceeded the monthly personal KPI and received a graduate full-time return offer.

Assistant Accountant Intern

Dec 2024 – Feb 2025

Online Air & Solar Pty Ltd, Melbourne, Australia

- Managed accounts-receivable processes in Xero and conducted project cost analysis using Excel.
- Presented cost-optimisation findings and redesigned internal training and onboarding documentation.

Business Analysis Intern

Jan – Feb 2023

FTLife Insurance Company Limited, Hong Kong (Remote)

- Conducted competitor, market, and KPI benchmarking analysis and presented findings to the project team.

Awards & Scholarships

Department of Applied Finance Prize

Macquarie University, 2023

Awarded as the highest achiever in AFIN2070 Stochastic Methods in Applied Finance after attaining 96/100 (HD). The distinction is officially recorded on the Macquarie University academic transcript.

Medibank Undergraduate Coursework Scholarship

Macquarie University, 2023

Awarded to a high-performing international undergraduate student based on academic merit and scholarship application assessment; eligibility required a minimum WAM of 80.

Recognition of High Academic Achievement

Macquarie University, 2023

Awarded in recognition of achieving a WAM above 80 and strong academic performance among residential students.

Jiangsu Provincial Government Scholarship for Overseas Study

Jiangsu, China, 2023

Provincial-level government scholarship awarded through university nomination and provincial review based on academic performance, disciplinary foundation, and development potential.

Vice-Chancellor's International Scholarship — Academic Excellence

Macquarie University, 2022

Competitive merit-based scholarship awarded to high-achieving international students upon admission.

Jiangsu University First-Class Scholarship

Jiangsu University, 2021, 2022

Awarded twice for ranking within the top 5% academically and satisfying broader performance and conduct requirements. Ranked first in the Finance major for four consecutive semesters.

Jiangsu University Outstanding Student Award

Jiangsu University, 2021, 2022

Awarded twice for outstanding academic performance, conduct, and well-rounded participation; recipients were limited to no more than 10% of eligible students within each faculty.

Research Methods & Technical Skills

Programming and data	Python (pandas, NumPy, scikit-learn), R, STATA, EViews
Machine learning	Regularised regression, hyperparameter comparison, rolling-window estimation, cross-sectional return prediction, out-of-sample model evaluation
Financial NLP	Large language model APIs, prompt engineering, information extraction, sentiment classification, textual-variable construction, LDA topic modelling
Financial econometrics	Panel regression, firm and time fixed effects, clustered standard errors, winsorisation, rolling CAPM beta estimation
Asset pricing and investment	Factor investing, characteristic-sorted portfolios, portfolio optimisation, risk-adjusted performance evaluation, Sharpe ratios
Financial operations	Transaction, position and portfolio reconciliation; break investigation; exception management; financial-data quality control
Languages	Mandarin Chinese (native); English (proficient)

References

Available upon request.